DAY16-WIRESHARK

THEORY (answer all 20 questions in your own words):

1. What is Wireshark?
   It is a Network Protocol Anayzer that inspects network packets and mostly used as essential for Network troubelshooting, protocol analysis, incident investigation, security monitoring and malware traffic analysis for isloated labs.\

2. What is packet capture?
   It is the process of recording network packets to analyze network communication.

3. What is a capture interface?
   It is a network connection could be Wi-Fi or ethernet that wireshark used to capture packets.

4. Name the three main Wireshark panes.
   - Packet List
   - Packet Detail
   - Packet Bytes

5. What is the difference between source and destination IP?
   Source IP: address sending the packet.
   Destination IP: address receiving it.

6. What does the Protocol column show?
   It shows protocols carried by each packet such as TCP, UDP, DNS or ICMP.

7. What is a Wireshark display filter?
   More like a searchbar but only shows specific packets that meet the conditions.

8. Which filter shows ICMP traffic: ICMP

9. Which filter shows DNS traffic: DNS

10. Which filter shows TCP traffic: TCP

11. What does tcp.port == 443 display?
    It displays TCP ports involving port 443, often used for HTTPS.

12. What is DNS?
    DNS stands for Domain Name System.
    It translates domain names into IP addresses.

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13. What is an ICMP Echo Request?
    It is a message sent by ping to check whether device is reachable.

14. What is an ICMP Echo Reply?
    It is the response sent by device when it receives ICMP echo request.

15. What is the TCP three-way handshake?
    It is a process used to establish TCP connection through message such as SYN, SYN-ACK, ACK.
16. What are SYN, SYN-ACK and ACK?
    SYN- Request a TCP connection.
    SYN-ACK – Acknowledges the request and responds it.
    SCK – Acknowledge the responds and complete the handshake.

17. Why is HTTPS content normally not visible as plain text?
    HTTP encrypts the data of application to normally prevent from unauthorized parties.

18. What is a PCAP/PCAPNG file?
    It is a file format used to save captured packets for later network analysis.

19. Why is Wireshark useful in incident response?
    It helps analyst investigate network traffic, understand possible incidents and unusual conditions.

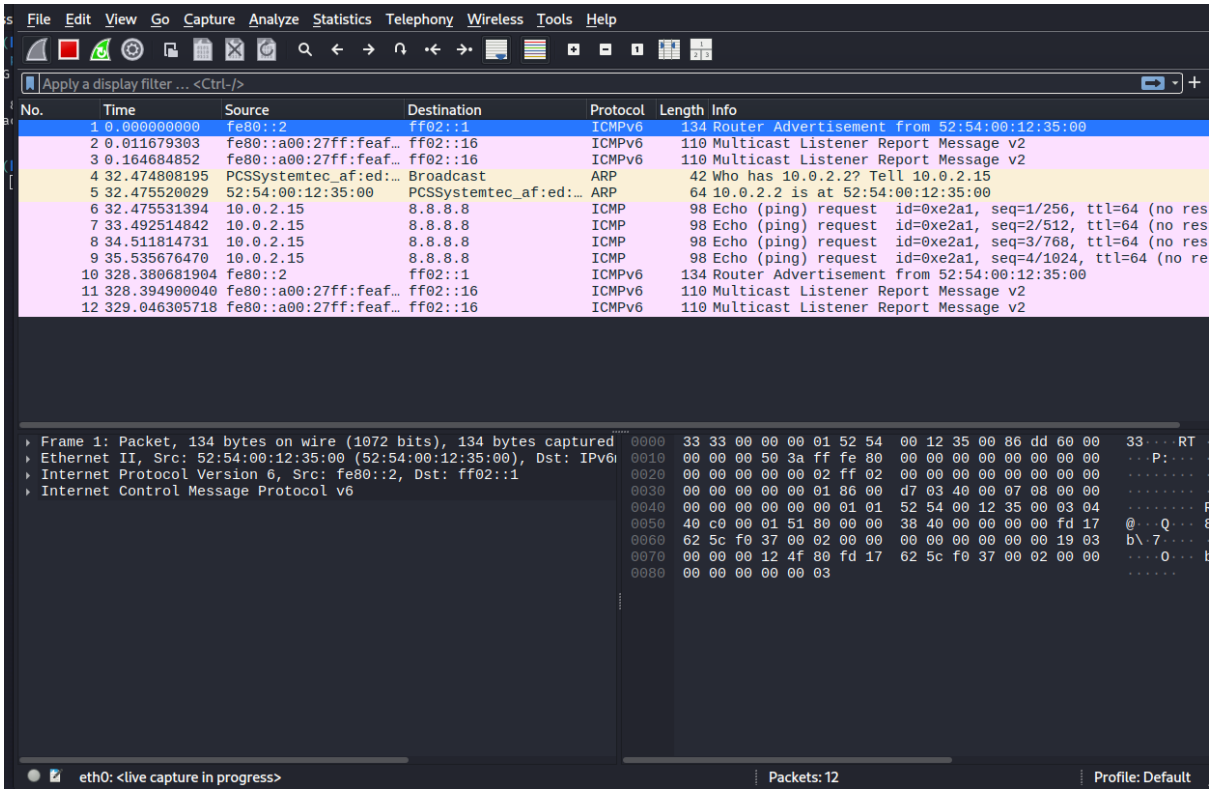
20. Why must packet capture be performed only on authorized traffic?
    Because capturing unauthorized traffic known as violating privacy, security laws or policies.

DAY 16 PRACTICAL LAB — Your First Wireshark Packet Analysis Lab

Capture ONLY your own device/VM/lab traffic.

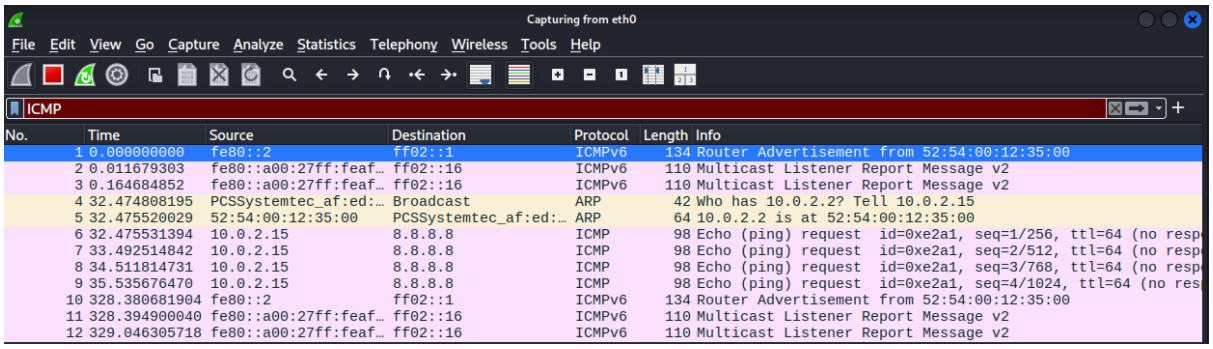
TASK 1 — Open Wireshark; identify your active interface; record its name.

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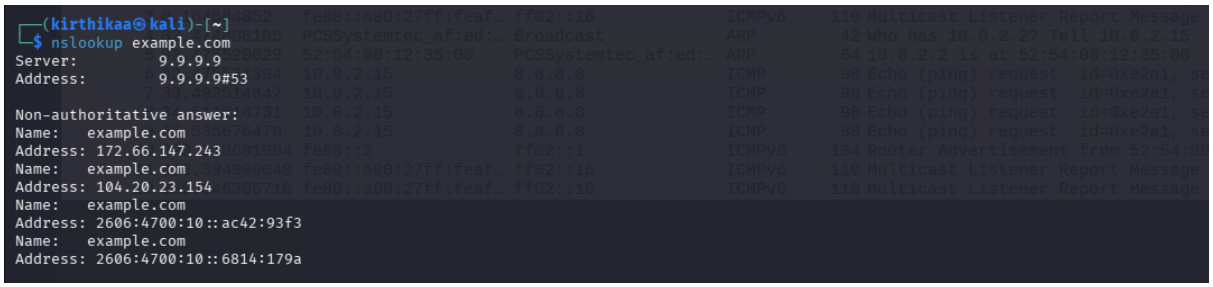


TASK 2 — Start capture ~20–30 seconds, then stop. Record roughly how many packets you captured.

TASK 3 — Fresh capture → run ping -c 4 8.8.8.8 (or ping 8.8.8.8 on Windows) → stop → filter 'icmp' → identify Echo Request, Echo Reply, Source IP, Destination IP.

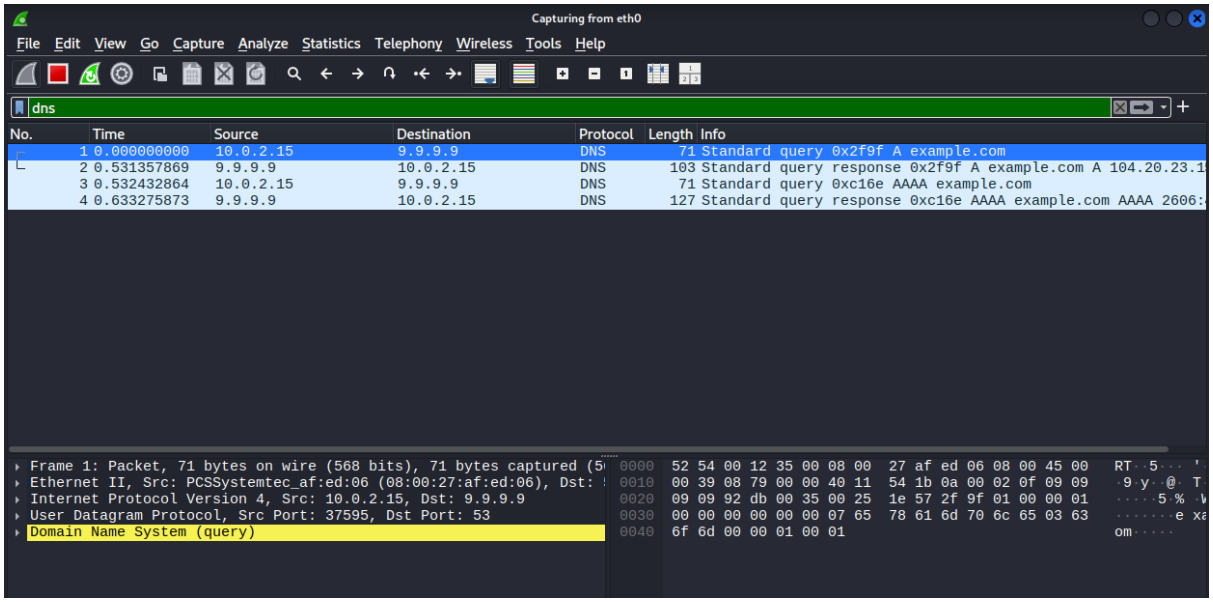


TASK 4 — Capture → run nslookup example.com (or getent hosts example.com) → stop → filter 'dns' → record the queried domain and returned address.

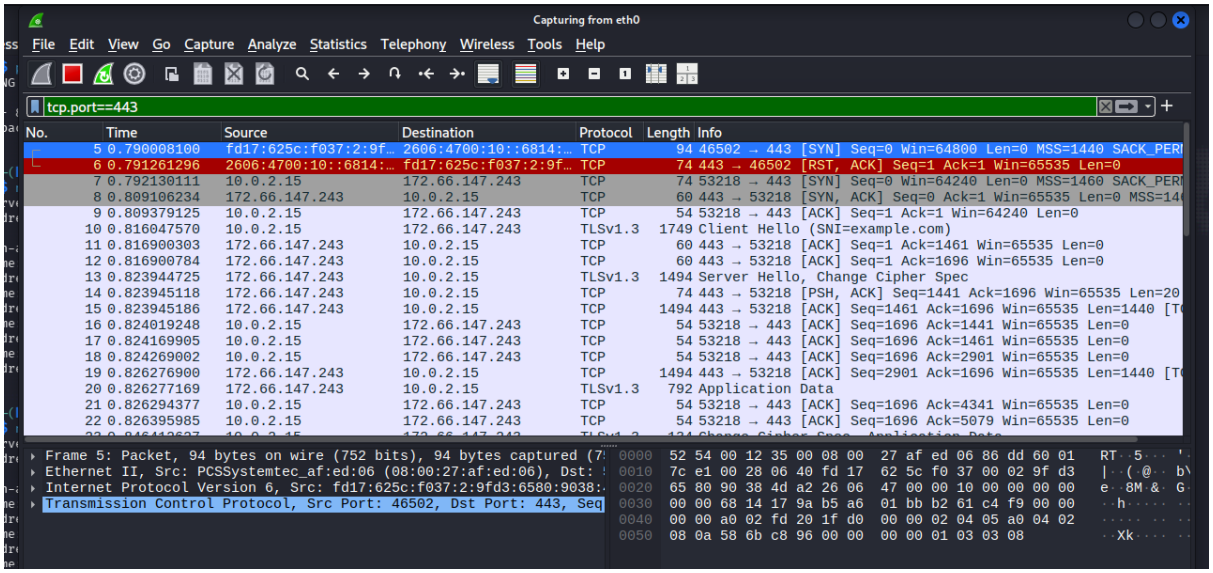


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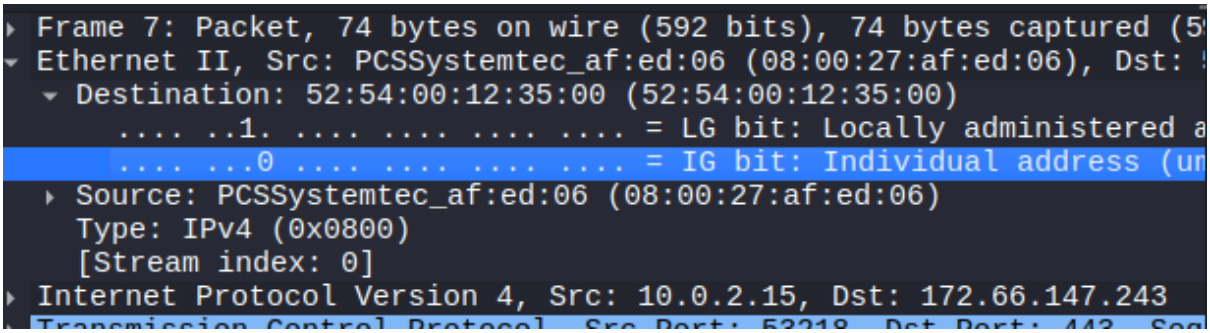
TASK 5 — Filter 'tcp' → pick one packet → record Source/Destination IP and Source/Destination port.



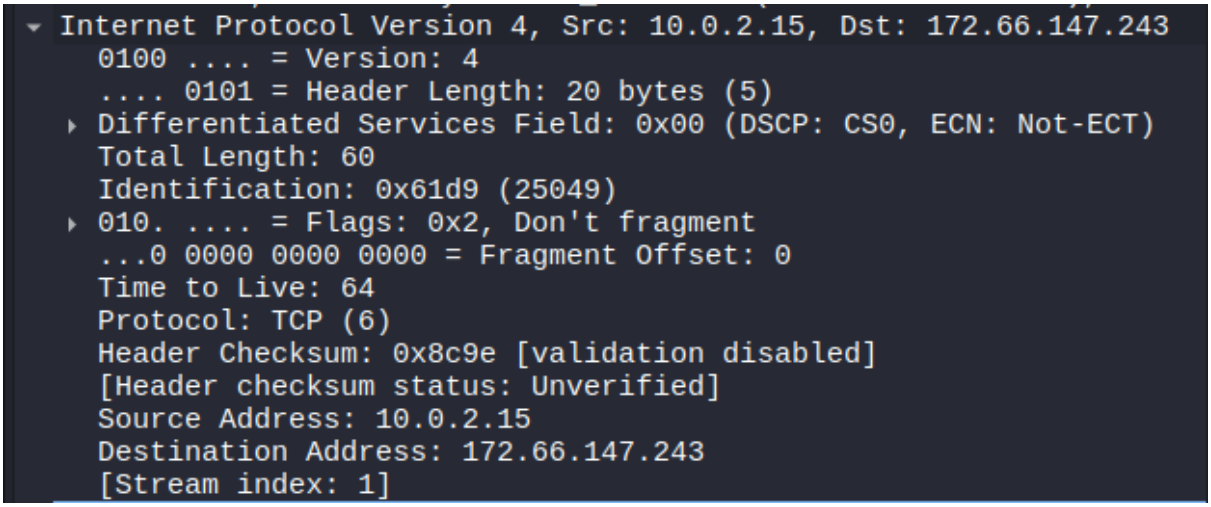
TASK 6 — Filter 'tcp.port == 443' → does HTTPS-related traffic exist in yours? If none: totally acceptable — do NOT generate sensitive traffic just to fill the screen.

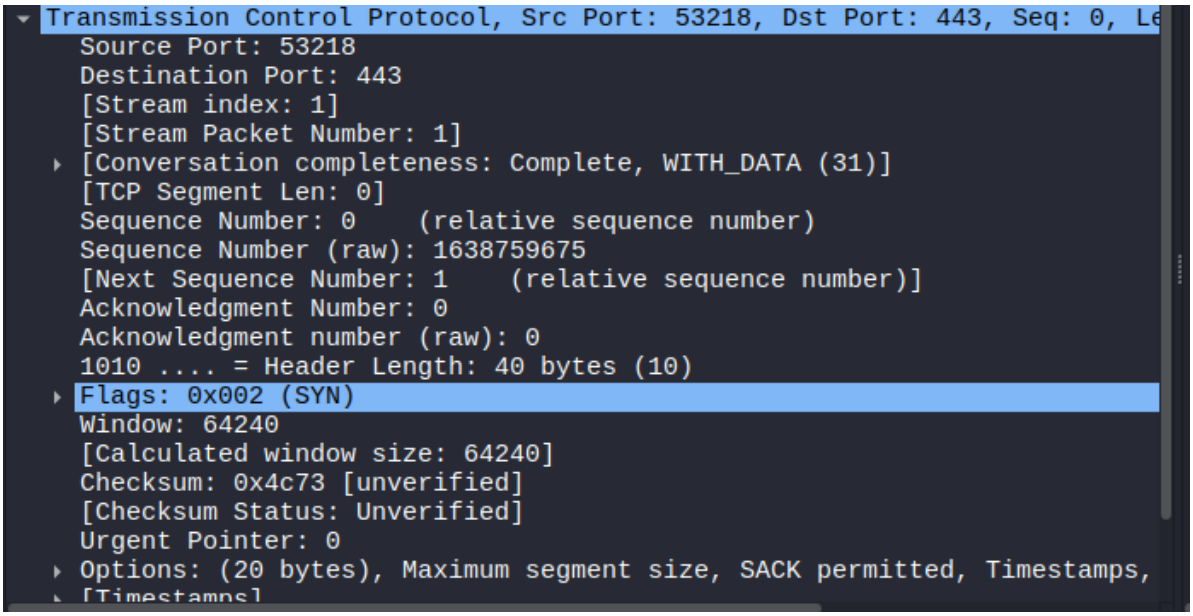


TASK 7 — Select one packet; expand Ethernet → Internet Protocol → TCP/UDP layers; record Source MAC, Destination MAC, Source IP, Destination IP, Protocol.



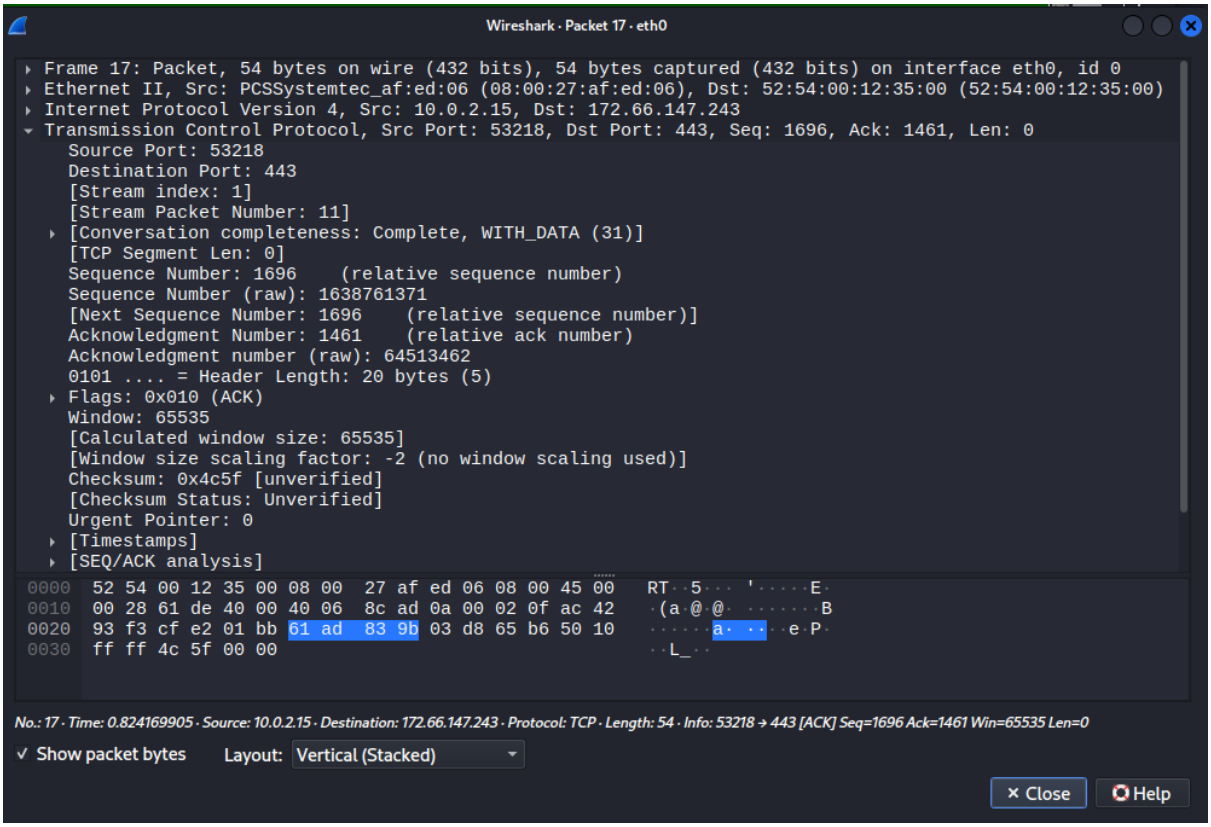
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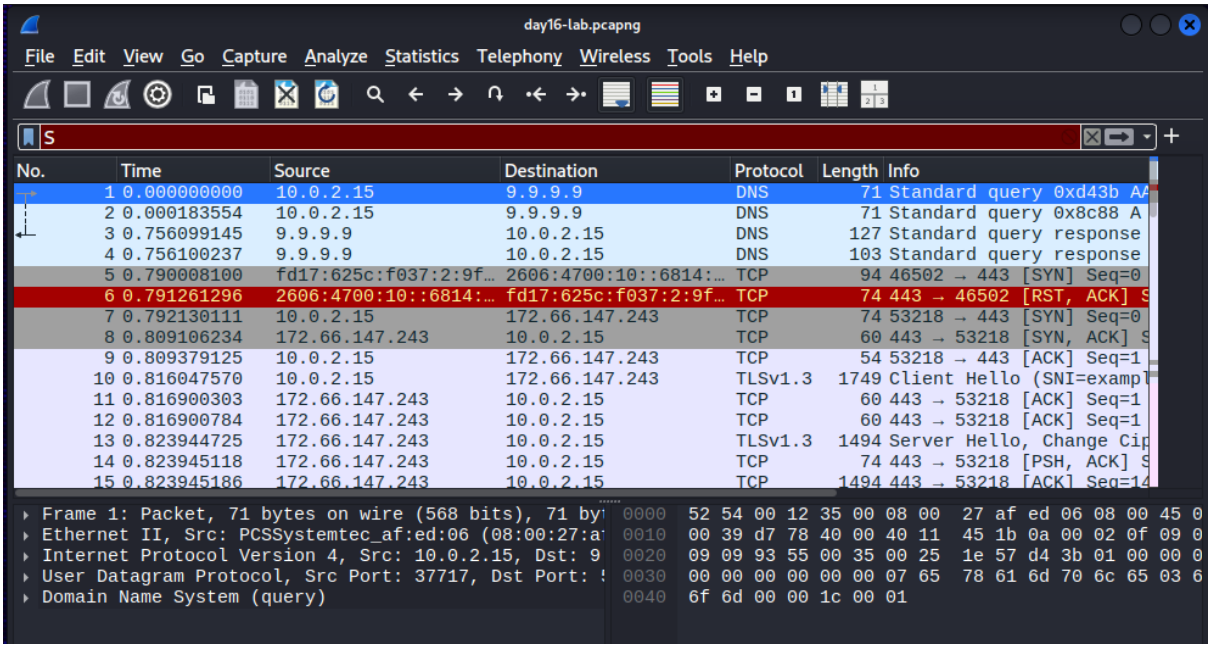


TASK 8 — If your capture has a handshake: identify SYN / SYN-ACK / ACK. If not: use the static example in Section 10 — no forced connections.

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TASK 9 — Save capture as day16-lab.pcapng; confirm the file exists.



TASK 10 — Mini investigation (static data, no capture needed):

P1: 192.168.1.10 → 8.8.8.8 ICMP

P2: 192.168.1.10 → 192.168.1.1 DNS

P3: 192.168.1.10 → 203.0.113.20 TCP dst-port 443

Which packet is ping-related? Which is DNS? Which is likely HTTPS-related TCP? (Answers: P1, P2, P3 — justify each from the columns.)

- P1, because ping commonly uses ICMP to test network connectivity.
- P2, because DNS resolves domain names into IP addresses.
- P3, because TCP port 443 is commonly used for HTTPS.